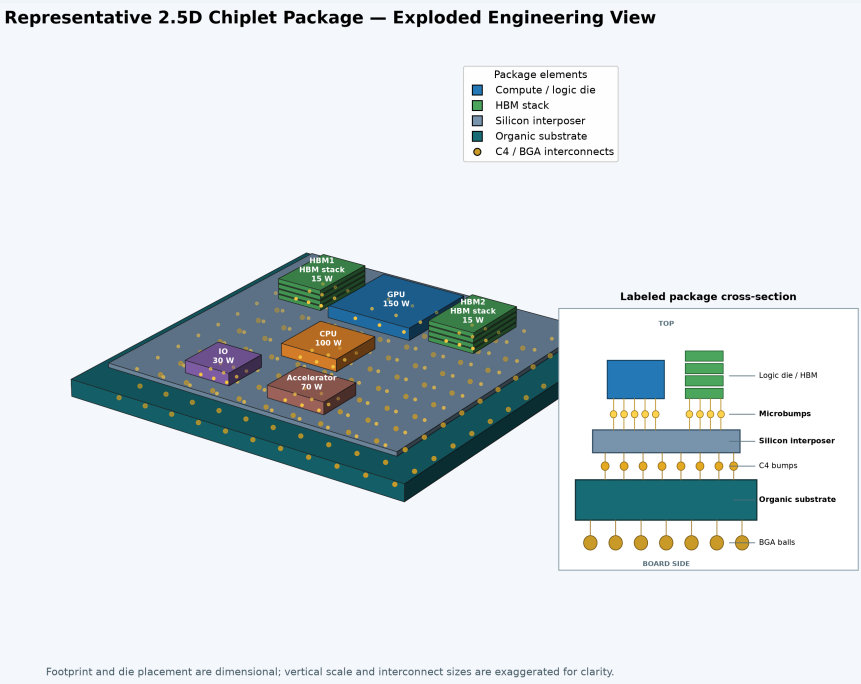


2.5D Chiplet Package Process Window

Reduced-order engineering exploration with transparent assumptions

Package architecture



Six chiplets on a silicon interposer and organic substrate. The model includes microbumps, C4 bumps, BGA balls, geometry validation, and weighted routing.

Baseline results

- 60.16 degC peak temperature
- 2.936 MPa maximum stress proxy
- 0.189 maximum manufacturing risk
- 215 weighted-mm routing metric
- 50 tests passing on Python 3.13.15

Engineering workflow

- 1 Validate package geometry and spacing
- 2 Estimate self heating and thermal coupling
- 3 Calculate the CTE mismatch stress proxy
- 4 Score five manufacturing-risk components
- 5 Evaluate a selectable five-factor DOE
- 6 Fit OLS, ANOVA, sensitivity, and interactions
- 7 Filter four-objective Pareto designs
- 8 Intersect five process-window constraints

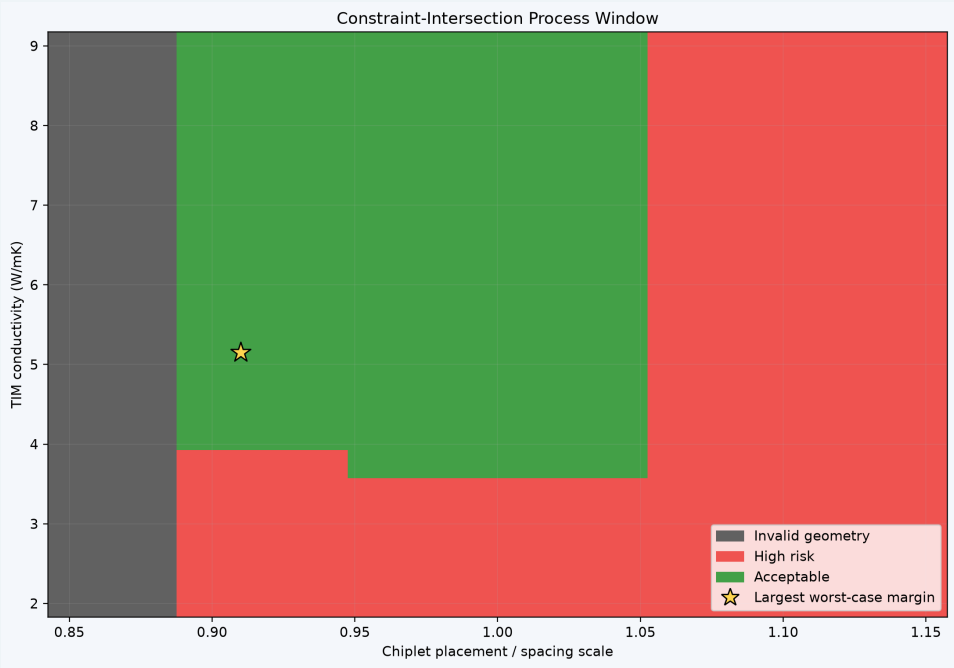
Key statistical finding

TIM conductivity accounts for about **59.5%** of the relative standardized main-effect sensitivity for peak temperature. Power scale contributes about **34.3%**. Residual non-normality limits classical p-value interpretation.

Model boundary

Comparative screening only. The thermal model is not FEA, the mechanical result is a proxy, and the thresholds are not OSAT qualification rules.

Process window



The 441-point map contains **172 acceptable**, 206 high-risk, and 63 invalid-geometry cases. Green points pass spacing, temperature, stress, risk, and routing limits together.

Portfolio deliverables

- Editable nine-tab Streamlit website
- JSON, YAML, CSV, PNG, SVG, and ZIP exports
- Technical report and implementation handbook
- Five-slide presentation and interview guide

Next OSAT step

Replace assumptions with a technology design kit, detailed solver models, measured process distributions, and physical test-vehicle correlation.